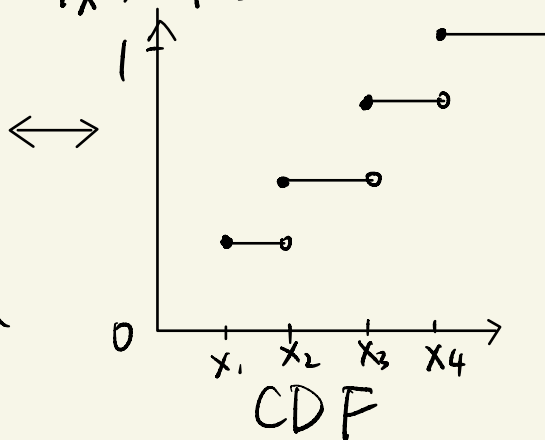
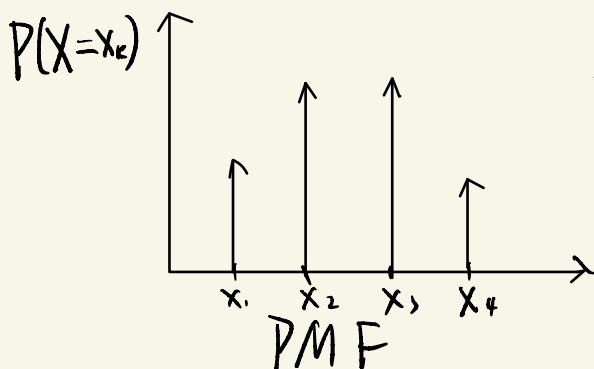


ECSE 4400
Lec 12
March 2nd

Topic: Probability Density Function (PDF) for continuous RV.

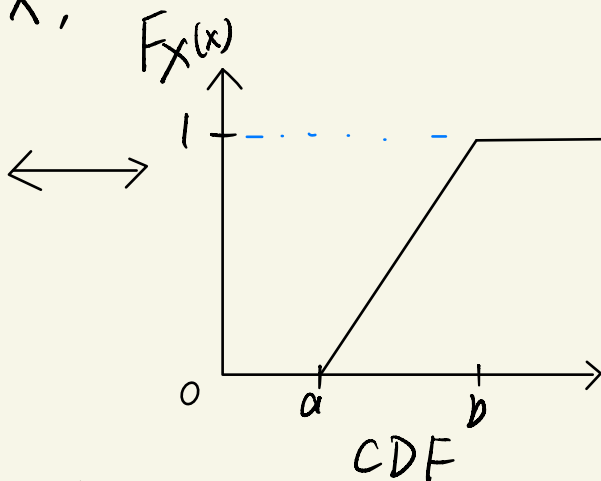
For discrete RV X ,

$$F_X(x) = P(X \leq x)$$



For continuous RV X ,

PDF



Q: What is the notion similar to PMF?

It is PDF, which is the most useful and important way of discussing a continuous random variable.

▷ Recall the definition of CDF (both discrete/continuous)

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x P_X(y) dy$$

In the discrete RV case, the function $P_X(y)$ is a impulse function, so $F_X(x)$ is a step function.

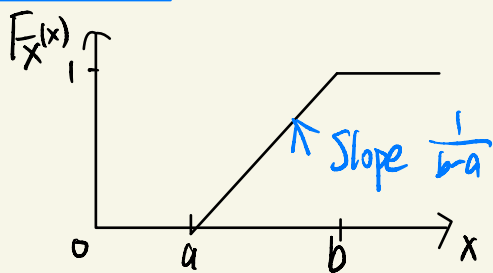
□ The PDF is defined as

$$f_X(x) = \frac{d}{dx} F_X(x)$$

↑ small case
big case

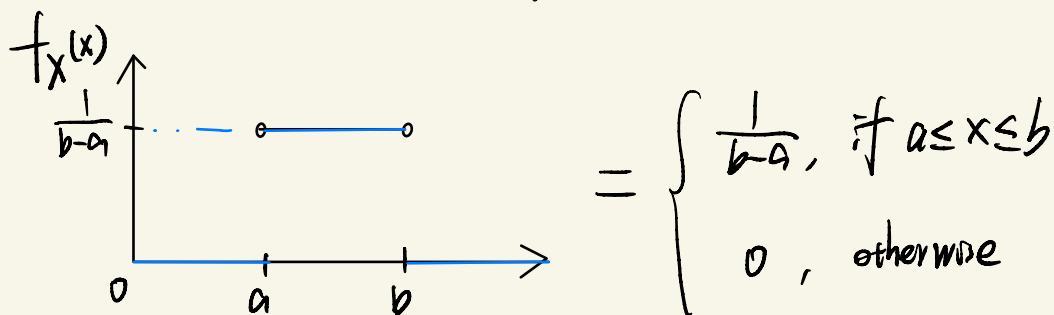
which makes sense since we got from the PMF to CDF via integration (summation).

Example - Uniform (Continuous) Random Variable



$$= \begin{cases} \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{if } x \leq a \\ 1 & \text{if } x \geq b \end{cases}$$

Therefore, the PDF of this RV is



At points $x=a$ and $x=b$, the derivatives are undefined, but it doesn't really matter since

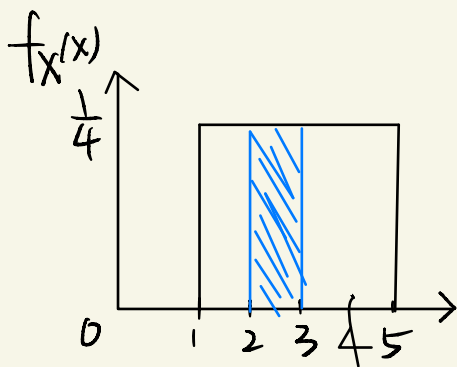
$P(X=a) = P(X=b) = 0$. Often we casually "connect the dots."

□ The usage of PDF? X is continuous RV

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

↑
if we know PDF of X

That is, the probability of X lying in an interval $[a, b]$ is the area under the interval,



$$\begin{aligned}
 P(2 \leq X \leq 3) &= \int_2^3 f_X(x) dx \\
 &= \int_2^3 \frac{1}{4} dx = \frac{1}{4}
 \end{aligned}$$

This can be also viewed from the fundamental theorem of calculus:

$$\begin{aligned}
 \int_a^b f_X(x) dx &= \int_{-\infty}^b f_X(x) dx - \int_{-\infty}^a f_X(x) dx \\
 &= F_X(b) - F_X(a) \\
 &= P(a \leq X \leq b)
 \end{aligned}$$

(Being a little sloppy about two endpoints)

Put another way,

$$\begin{aligned}
 P(x < X \leq x+h) &= F_X(x+h) - F_X(x) \\
 &= \frac{F_X(x+h) - F_X(x)}{h} \cdot h \stackrel{h \rightarrow 0}{=} f'_X(x) \cdot h \\
 &= f_X(x) \cdot h
 \end{aligned}$$

$\uparrow$
 a small constant

Before we give more examples, one more thing I want all of you to be careful is that

Notice: PDF $f_X(x)$ is not the "Probability of X getting the value x !"

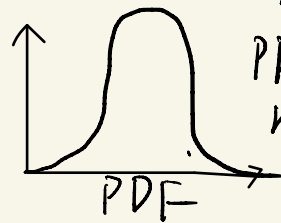
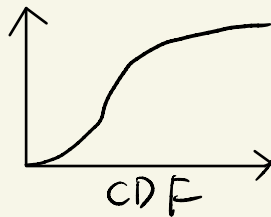
Why? The probability of continuous RV getting any particular value x is 0 !

What it is? The idea of PDF describes the "density" of probability near the value x .

If the PDF is high near x , we expect to get more outcome around this area!

□ Properties of PDF

1) $f_X(x) \geq 0$ for all x

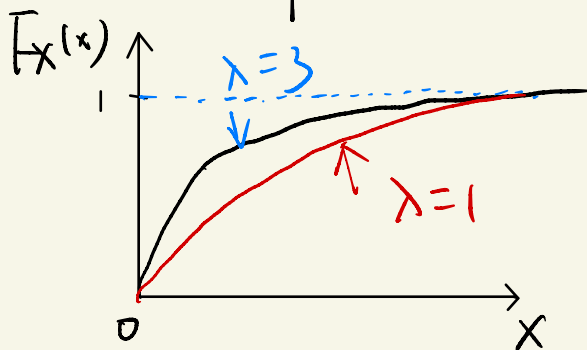


CDF is non-decreasing;
PDF is nonnegative.

$$2) \int_{-\infty}^{\infty} f_X(x) dx = 1, \quad = F_X(\infty) = 1$$

□ Some new examples of continuous RVs

1) The **exponential** random variable has CDF



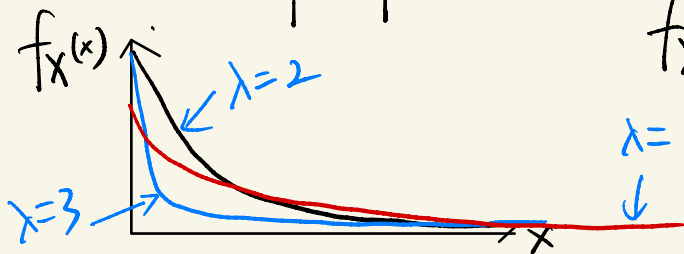
$$F_X(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

As parameter λ **decreases**, the CDF becomes close to linear.

Q: What real-world examples the RV can model?

– How long does a device last?

The PDF of exponential RV



$$f_X(x) = \begin{cases} 0, & x < 0 \\ \lambda e^{-\lambda x}, & x \geq 0 \end{cases}$$

↑
positive constant

— Time between two school buses arrive

Additional discussion about Exponential RV.

We often care about events of the form

$$P(X > x)$$

e.g., Probability devices last longer than x years.

$$P(X > x) = 1 - P(X \leq x)$$

Plug in the $\Rightarrow 1 - F_X(x)$

CDF of exponential RV $\searrow = e^{-\lambda x}, \quad x \geq 0$

A connection between the Poisson Discrete Random variable and the exponential random variable.

Let N (integer) denote the number of events happen in t seconds (e.g., packet arrivals), where the Poisson parameter $\alpha = \lambda t$ (λ is the average arrival rate in packets/second).

Q: What is the probability that the next arrival occurs t or more seconds later?

Equivalently, $P(N=0) \underset{\substack{\uparrow \\ \text{PMF of Poisson} \\ \text{RV}}}{=} \frac{\lambda^0}{0!} e^{-\lambda} = e^{-\lambda} = e^{-\lambda t} = P(X > t)$

where X denotes the time until the next packet arrival.

That is, $N=0$ and $X > t$ are equivalent events. One looks at it as # of packets and the other as the time to packet arrival.

□ Other examples of continuous RVs.

i) Gamma RV

$$f_X(x) = \frac{\lambda (\lambda x)^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, \quad \alpha < \infty$$

with

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx.$$

2) Beta RV

$$f_X(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}, 0 < x < 1$$